

Introduction to Machine Learning (ML)

What is Machine Learning?

Machine Learning is a subset of **Artificial Intelligence (AI)** that allows systems to learn from data, identify patterns, and make decisions with minimal human intervention.

Rather than being explicitly programmed for every task, a machine learning model is trained on data and then makes predictions or decisions based on what it has learned.

Formal Definition:

Machine Learning is the study of computer algorithms that improve automatically through experience. — **Tom M. Mitchell**

Why Do We Need Machine Learning?

Traditional programming cannot handle tasks where:

-  Rules are too complex to write explicitly (e.g., face recognition).
-  Data is too large or varied for manual analysis (e.g., analyzing social media trends).
-  Environments change over time (e.g., stock market prediction, weather forecasting).

Key Benefits:

-  Automation of repetitive or complex tasks.
 -  Adaptability to new data or changing conditions.
 -  Insight extraction from big data.
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Types of Machine Learning

Type	Description	Examples
Supervised Learning	The model learns from labeled data (input → output pairs).	Spam detection, loan approval, image classification
Unsupervised Learning	The model finds patterns in unlabeled data.	Market segmentation, topic modeling, anomaly detection
Reinforcement Learning	The model learns by trial and error, receiving rewards or penalties.	Robotics, game playing (e.g., AlphaGo), self-driving cars

How Machine Learning Works — Basic Workflow

1. **Data Collection**

Gather relevant and high-quality data.

- 2. **Data Preprocessing** ✍️
Clean, transform, and prepare data for training.
- 3. **Model Selection** 🔍
Choose an appropriate algorithm (e.g., decision trees, neural networks).
- 4. **Training** 🧑‍🔬
Feed data into the model so it can learn patterns.
- 5. **Testing and Evaluation** 📊
Test the model's performance on unseen data.
- 6. **Deployment** 🚀
Use the trained model in real-world applications.
- 7. **Monitoring & Updating** 🔄
Continuously monitor and improve the model as needed.

📁 Common Machine Learning Algorithms

Algorithm	Type	Use Case
Linear Regression	Supervised (Regression)	Predicting housing prices
Decision Trees	Supervised (Classification/Regression)	Loan approval, diagnostics
K-Means Clustering	Unsupervised	Customer segmentation
Support Vector Machines (SVM)	Supervised	Image recognition, text classification
Neural Networks (Deep Learning)	Supervised/Unsupervised	Speech recognition, self-driving cars

⚙️ Real-World Applications

- 🏥 **Healthcare**: Disease diagnosis, drug discovery.
- 💰 **Finance**: Fraud detection, algorithmic trading.
- 🛒 **E-commerce**: Product recommendation systems.
- 🗣️ **Natural Language Processing**: Chatbots, language translation.
- 🌾 **Agriculture**: Crop monitoring, yield prediction.

🔗 Relation to Your Course (CS3151)

According to your course outline:

- **Week 12**: Introduction, Need, and Types of Learning
- **Week 13**: Supervised Learning – Classification and Regression
- **Week 14**: Neural Networks Basic Concepts

You will transition from understanding basic ML principles to exploring real-world ML models and how they are applied in AI systems.

By the end of this module, you should be able to **describe, design, and implement** simple machine learning models for problem-solving.

Recommended Reading

-  *Artificial Intelligence: A Modern Approach* by Russell & Norvig (**Chapters 19 & 20**).
-  *Pattern Classification* by Duda, Hart, and Stork.